ANSWERS

***High** voltage pn diodes.*

a) [bookwork] Explain how and when avalanche breakdown occurs in a diode, and its effects. *[5]*

Avalanche breakdown occurs due to impact ionisation under a **high electric field** [1], typical in a depletion region in reverse bias. Charge carriers are accelerated under this electric field and will undergo scattering processes, **impacting the atoms of the lattice** [1]. If the **kinetic energy transferred from the carriers to the lattice** [1] is larger than the energy of the bandgap, **electron-hole pairs will be generated** [1] which in turn will be accelerated, by the high field gaining kinetic energy before again impacting and in turn **producing more carriers in a chain reaction**. This leads to rapid increase in currents, with the **risk of thermal runaway/overheating** [1].

b) [application of knowledge and calculation], The maximum electric field, E_{max}, within a diode as a function of the voltage, V, across the diode can be derived from Poisson's equation as:

$$E_{max} = \sqrt{\frac{2eN_AN_D(V_0 - V)}{\varepsilon(N_A+N_D)}}$$

where V_o is the built-in voltage. Where in the device is this field a maximum? Show that the for a p^+n diode the reverse voltage is limited by the avalanche breakdown field, E_{BD}, to a maximum of approximately

$$V \approx -\frac{\varepsilon E_{BD}^2}{2eN_A}$$

and calculate the doping level to allow at least a 100 V reverse voltage. Is the actual voltage likely to be larger or smaller, and why? *[11]*

The field will be a maximum at the np **interface/junction**. [1]

Rearranging

$$E_{max} = \sqrt{\frac{2eN_AN_D(V_0 - V)}{\varepsilon(N_A+N_D)}}$$

$$V = V_0 - \frac{\varepsilon(N_A+N_D)}{2eN_AN_D}E_{max}^2$$ [1]

For a p^+n diode, $N_A >> N_D$ and V_o is less than 1 V so [1]

$$\frac{N_A+N_D}{N_AN_D} \approx \frac{1}{N_D}$$ [1]

Substituting for $E_{max} = E_{BD}$, and assuming that the breakdown voltage is much greater than the built-in voltage of around 0.7 V gives

$$V \approx -\frac{\varepsilon}{2eN_D}E_{BD}^2$$ [1]

The doping density for a reverse breakdown voltage of 100 V is given by:

$$N_D \approx \frac{\varepsilon}{2eV}E_{BD}^2$$ [1]

$N_D \sim 11 \times 8.85\text{E-}12 \times (3\text{E}7)^2/(2 \times 1.6\text{E-}19 \times 100)$ [1]

$N_D \sim 2.7E15\ /cm^3$ [1]

Likely **smaller** [1] as the formula does not take into account any **nonuniformities in the field [1]**, with any local field enhancement causing a **lower onset of avalanche breakdown [1]** that will initiate a chain reaction.

c) [bookwork] Explain why for a p^+n diode used in a switching power application a lightly doped, but not intrinsic, n^- layer is introduced. *[5]*

For power electronics, the diode needs to be able to **maintain a large reverse bias voltage** [1] and thus needs to **avoid avalanche breakdown** [1]. Hence a **lightly doped layer will reduce the breakdown voltage** [1]. A lightly doped, rather than an intrinsic layer is preferred as the **on-resistance of the diode is lower** [1] and the **shorter lifetime of the minority carriers due to dopant scattering means faster switching** [1]. It also **increases the punch-though voltage** [1] (only two out of these three scored).

d) [bookwork and application of knowledge] Explain "punch through" and how the width of the n^- layer sets the reverse diode voltage at which it occurs? *[4]*

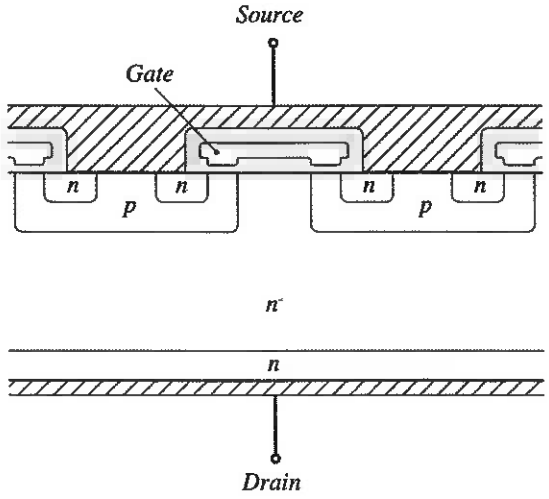
The n- layer will be **increasingly depleted with reverse bias across the diode** [1], with the **depleted region increasing into the n^- layer from the p^+n^- junction** [1]. At punch though the **depletion region extends through the entire n^- layer** [1]. **Increasing the layer width will increase the reverse voltage** at which this occurs [1].

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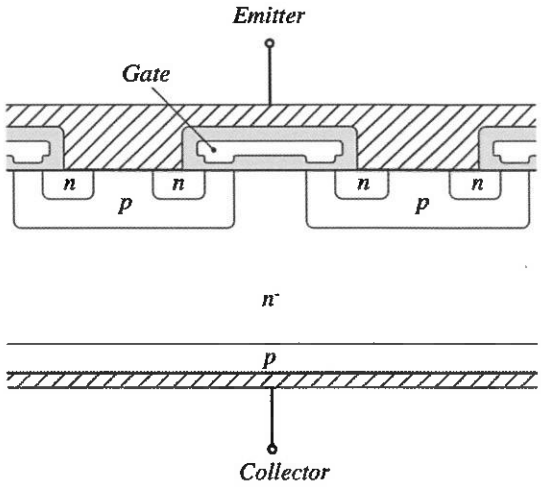
2. A cross section of a generic, vertical-geometry power device is shown in fig. 2.1, with the critical device regions and contacts outlined but unlabelled.

a) [bookwork] Draw two copies of this cross section and label all the contacts and the doping of each device region for i) a DMOS (diffused metal-on-insulator), ii) an IGBT (insulated gate bipolar transistor) power configuration. [5]

DMOS



IGBT



Subtract 1 mark for each incorrect labelling [5]

b) [bookwork] For each of these two devices, explain how the current through the device is controlled. [5]

DMOS: the current is **controlled by the voltage of the gate** [1].When the device is on, the **channel is inverted and the electron current though the device** flows from the n source through the n-channel under the gate to the n^- drift region and down to the n drain [1]. When the **gate is below the threshold voltage for the channel is not inverted and little current flows.**

IGBT: the current is **controlled by the voltage of the gate** [1]. When the device is on, the **channel is inverted and the electron current provides a base current for the vertical pn⁻p BJT** [1]. When the **gate is below the threshold voltage for the channel is not inverted and no base current flows** [1].

c) [bookwork, and application in making comparison] For the DMOS, what predominantly determines the ON resistance, and how does this change in the IGBT? [5]

DMOS: The on resistance is dominated by the **JFET resistance due to the depletion of the narrow n- region below the gate** [1] and the **drift region resistance in the bulk n- region due to the low doping** [1].

IGBT: The collector current through the drift region now includes **minority injection** from the collector [1]. This **increases the carrier density, reducing the on resistance** [1]. In addition, at **high-level injection** the **majority carrier concentration further increases** with voltage, further reducing the on resistance [1].

d) [bookwork, and application in making comparison] For each of these two devices, what limits the switching speed, and which device is faster and why? [5]

DMOS: as a majority device, **the gate capacitance and speed are the speed limits to switching the n-channel** [1]. **These depend on the insulator thickness, and the charge state of the channel** [1].

IGBT: The speed is set by **the switching of the base current by the gate** [1], similar to the DMOS, and the **storage delay as the minority carriers are removed from the n^- base** region [1].

Overall, the **DMOS is faster** [1] as it **does not have the need to discharge the base region during switching** [1].

e) *[Application of knowledge] In the light of above considerations, which of the two devices would be more appropriate for use in the battery power management of an electric vehicle, and identify one further device parameter that will influence the choice.* *[5]*

The **on resistance** [1] will likely be the most important parameter, as the power management should **not itself consume much power through resistive heating** [1]. **Switching speed is likely a secondary consideration** [1]. **This suggests an IGBT is the better choice** [1]. Further considerations that might influence the choice might be i) the reverse leakage current, ii) the cost, iii) the maximum current rating [1 for any]